

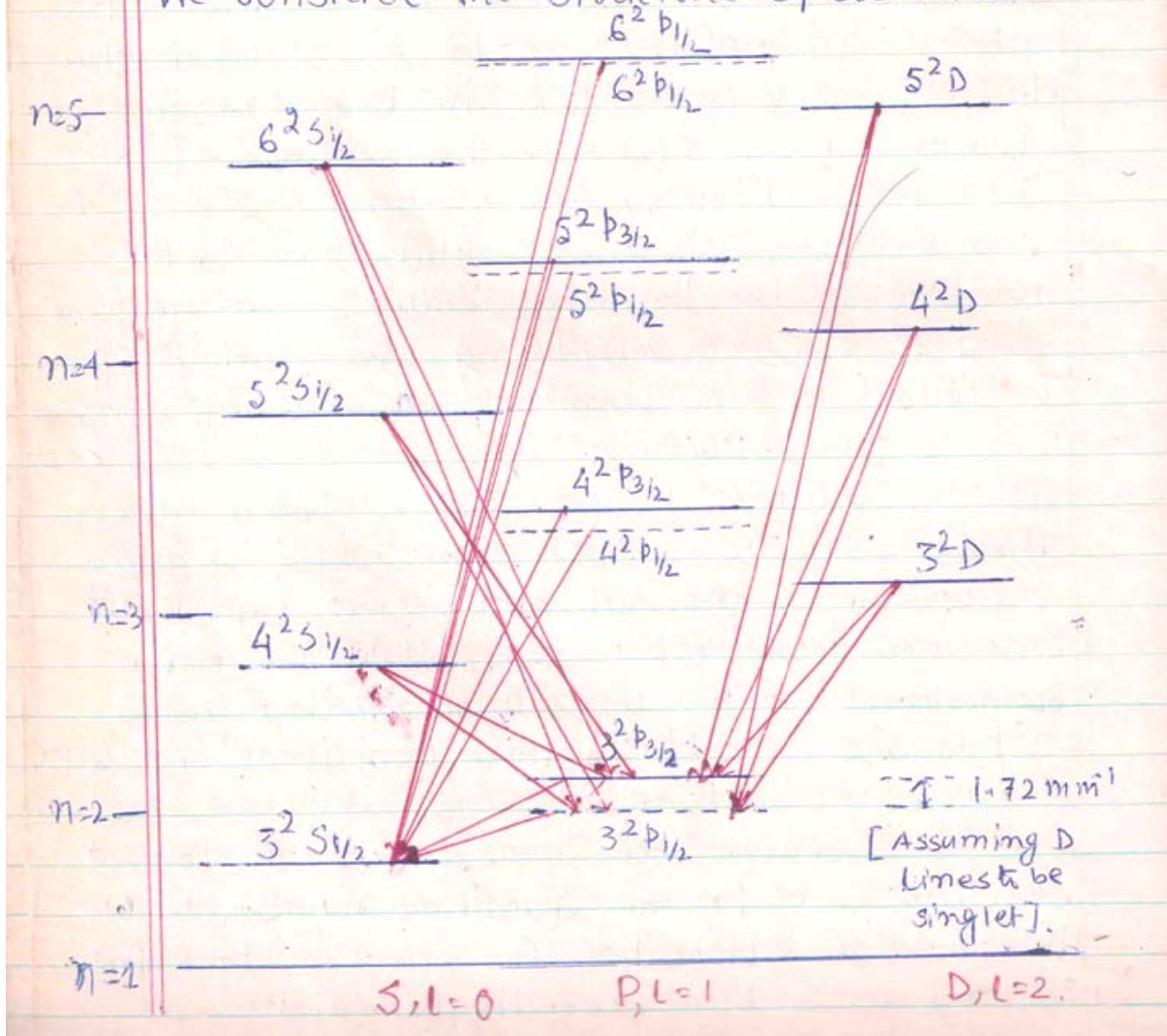


Fine Structure

Fine Structure and Electron Spin

It is known that the 1st member of the principal series is the Sodium yellow line which with quite moderate resolving power is seen to consist of two lines separated by about 0.6 nanometre. It is also found that all the lines in the sharp, principal and diffused series are doublets, so that they are said to have fine structure.

We consider the structure of sodium lines:





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We get the following table from experimental observations:

Sharp Series $\lambda(\text{mm})$ $\bar{\sigma}(\text{mm}^{-1})$ $\Delta\bar{\sigma}$	Principal $\lambda(\text{mm})$ $\bar{\sigma}(\text{mm}^{-1})$ $\Delta\bar{\sigma}$	Diffuse Series $\lambda(\text{mm})$ $\bar{\sigma}(\text{mm}^{-1})$ $\Delta\bar{\sigma}$
1140.42 876.63 1138.24 878.31	588.996 1697.39 589.593 1695.62	819.482 1219.95 818.330 1221.66
616.072 1622.74 615.421 1624.45		
515.364 1939.83 514.909 1941.55		
475.189 2103.84 474.802 2105.56		
454.522 2199.50 454.167 2201.22		
442.331 2260.12 441.994 2261.84		

From the experimental measurement of wave length of the doublets it is found that the wave number separation ($\Delta\bar{\sigma}$) of each pair of doublets is same equal to 1.72 mm^{-1} in the Sharp and Diffuse Series. In the principal series ($\Delta\bar{\sigma}$) is not constant and becomes smaller as we proceed to higher members of the Series. However, the doublets of the 1st member of the principal series is also 1.72 mm^{-1} .



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In terms of energy level diagram for sodium, since both the Sharp and Diffuse Series involve 3P term we can imagine that this 3P term consists of two levels with a separation of 1.72 mm^{-1} . This explains the constant wave number separation of the lines in both Sharp and Diffuse Series assuming both the 's' & 'd' levels to be single (As shown in the energy level diagram). However the D-levels are not always single. To explain how the doublets of the principal series close up as we proceed up the series we must suppose that all the P-terms are doubled and that the separation becomes smaller for the higher terms.

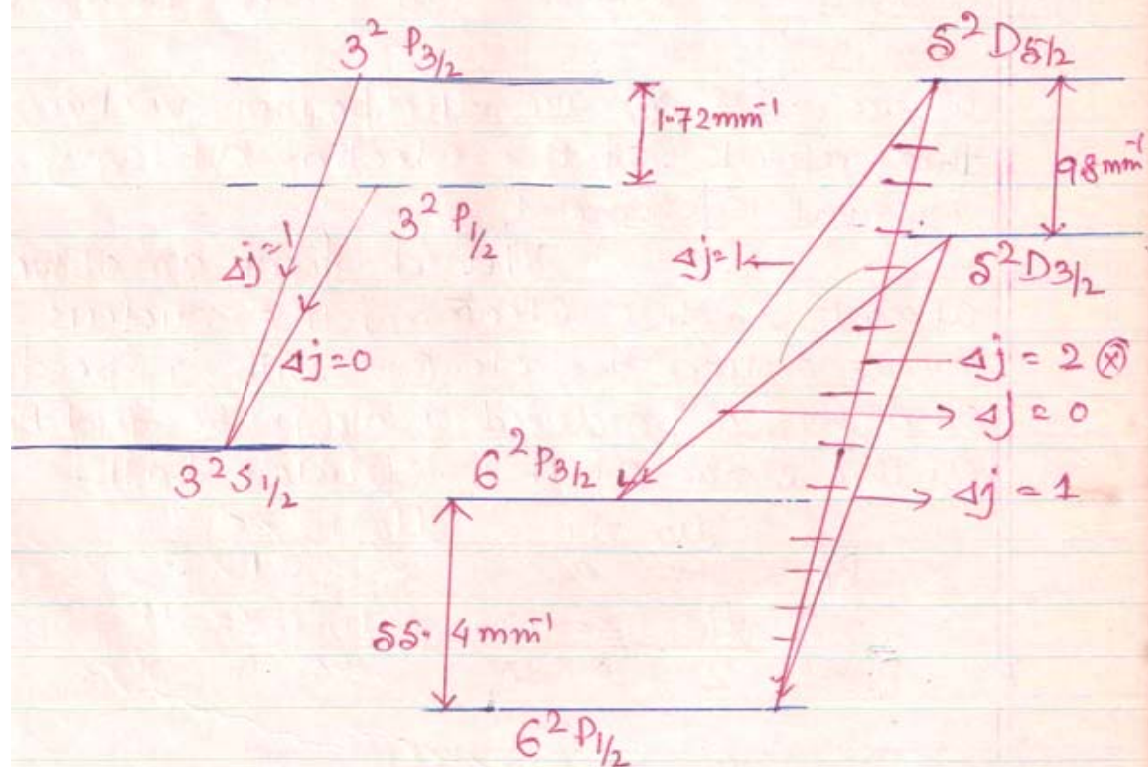
The fine structure of the spectral lines can be explained by the introduction of electron spin. To explain the fine structure we have assumed above the term separation of the l-levels which is not very apparent, but it is obvious that some further quantum restriction is needed. Since all the principal series lines are doublet and not quadruplet in sodium D-lines we conclude that s-levels can not be doublets where as all other levels are doublets. It is found empirically that this difficulty could be resolved by adding algebraically quantum number $1/2$ of unknown origin, to the orbital quantum no. l to give two resultant values of total quantum number j given by $j = l + \frac{1}{2}, l - \frac{1}{2}$. This gives two



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different levels for each 'L', except $L=0$, because j values of 'S' levels can only be $+\frac{1}{2}$ but not $-\frac{1}{2}$.

If we now redraw the energy level diagram for two sodium D-lines and the cesium $6P-5D$ diffuse triplet in terms of new j quantum number as shown in the figure below we see that the doublets and triplets can arise only when Δj is limited to 0 or 1 but not 2 i.e. $\Delta j = \pm 1$ and 0.



In 1925 Goudsmit and Uhlenbeck postulated the hypothesis of spinning electron with an associated angular momentum of $\frac{1}{2}h$ and a quantum number of $S = \frac{1}{2}$. The coupling of orbital and the spin angular momentum - govern by the Quantum number rule $j = l + \frac{1}{2}$ and $l - \frac{1}{2}$.



Spin Orbit Interaction and the fine Structure Of the Spectral Lines of Hydrogen:-

The Orbital angular momentum of the single electron (valence electron) moving under a central force due to the nuclear charge (Ze) of an hydrogen like atom is :-

$$\vec{L} = \vec{r} \times \vec{p} \quad \text{where } \vec{p} \text{ is linear momentum.}$$

$$\text{or } \vec{L} = \vec{r} \times m\vec{v} = m \vec{r} \times \vec{v} = L^* \hbar \quad \text{--- (1)}$$

where $\vec{r}$ & $\vec{v}$ are the position vector & the velocity of the electron at any instant of time t .

Viewed from an observer at rest in the electron; the nucleus moves round the electron with a speed $\vec{v}$ and hence produced a magnetic field $\vec{B}_l$ at the centre of the circular path.

$$\vec{B}_l = \frac{\mu_0}{2} \frac{qi}{a} = \frac{\mu_0}{2} \frac{1}{\left(\frac{T}{r}\right)} \frac{Ze}{r}$$

$$\vec{B}_l = \frac{\mu_0}{2} \frac{Ze}{r} \frac{v}{2\pi} = \frac{\mu_0}{4\pi} \frac{Ze}{r} \frac{v}{r}$$

$$\vec{B}_l = \frac{\mu_0}{4\pi} \frac{Ze}{r^3} (\vec{r} \times \vec{v})$$

Putting the value of $\vec{r} \times \vec{v}$ from (1) :-

$$\vec{B}_l = \frac{\mu_0}{4\pi} \frac{Ze}{mr^3} L^* \hbar \quad \text{--- (2)}$$

The spinning electron having a spin



magnetic moment $\vec{\mu}_s$ in the magnetic field of induction vector $\vec{B}_L$ experiences a torque or couple :-

$$\vec{T} = \vec{\mu}_s \times \vec{B}_L$$

$$\left[\begin{aligned} T &= \mu B \sin \theta \\ &= \vec{\mu} \times \vec{B} \end{aligned} \right]$$

and due to this torque the angular momentum vector of the electron changes. But since torque $\vec{T}$ is perpendicular to both the vectors $\vec{B}_L$ and $\vec{\mu}_s$ i.e. $\vec{P}_s$, the magnitude of angular momentum vector does not change but the direction changes i.e. the angular momentum vector precesses around the direction of $\vec{B}_L$ and is known as Larmor's precession.

It can be shown that the angular velocity of Larmor's precession

$$\vec{\omega} = \frac{e}{2m} \vec{B}_L \quad [\text{normal Zeeman effect :- proof}]$$

$$\text{or } \vec{\omega} = \frac{e}{2m} \cdot \frac{\mu_0}{4\pi} \frac{Ze^2 \hbar}{m r^3} l^* = \frac{\mu_0 Ze^2 \hbar}{8\pi m^2 r^3} l^* \quad \text{--- (3)}$$

Due to this precessional motion, the energy changes & the change in energy is given by:

$$\Delta W_{L,S} = \vec{P}_s \cdot \vec{\omega} = s^* \hbar \cdot \frac{\mu_0 Ze^2 \hbar}{8\pi m^2 r^3} l^*$$

$$\text{or } \Delta W_{L,S} = \frac{\mu_0 Ze^2 \hbar^2}{8\pi m^2 r^3} l^* \cdot s^*$$

$$\text{or } \Delta W_{L,S} = \frac{\mu_0 Ze^2 \hbar^2}{8\pi m^2 r^3} l^* s^* \cos l^* \wedge s^*$$



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The torque acts normal to the plane containing the vectors $\vec{L}$ and $\vec{S}$ and therefore $\perp$ to their resultant $\vec{J}$.

According to conservation of angular momentum the total angular momentum $\vec{J}$ of the electron should remain constant both in magnitude and direction. This is satisfied if the spin orbit interaction leads to a precession of $\vec{L}$ and $\vec{S}$ round their resultant $\vec{J}$ keeping the $\vec{L} \cdot \vec{S}$ constant.

Using properties of triangle:-

$$\cos \angle L \hat{S} = \frac{L^2 + S^2 - J^2}{2LS}$$

$$\text{or } L S \cos \angle L \hat{S} = \frac{L^2 + S^2 - J^2}{2} \quad \text{--- (5)}$$

Putting equation (5) in (4) :-

$$\Delta W_{l,s} = \frac{M_0 Z e^2 \hbar^2}{8\pi m^2 r^3} \left[\frac{L^2 + S^2 - J^2}{2} \right] \quad \text{--- (6)}$$

Since interaction energy $\Delta W_{l,s}$ is small compared to the total energy of the orbit hence we can calculate $\Delta W_{l,s}$ by taking the time average $\left(\frac{1}{r^3} \right)$ of



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the only variable $\frac{1}{r^3}$ of the electron in the orbit.

It can be shown that :-

$$\left(\frac{1}{r^3}\right) = \frac{z^3}{a_0^3 n^3 l(l+\frac{1}{2})(l+1)} \quad \text{--- (7)}$$

where a_0 = Radius of the 1st Bohr orbit.
Putting equation (7) in (6) and dividing by hc ; we get the interaction energy in terms of wave number

$$\pi = \frac{\Delta W_{LS}}{hc} = \frac{a [l^{*2} + s^{*2} - j^{*2}]}{2} \quad \text{--- (8)}$$

$$\text{Where } a = \frac{\mu_0}{8\pi} \frac{ze^2 \hbar^2}{m^2} \frac{z^3}{a_0^3 n^3 l(l+\frac{1}{2})(l+1)} \cdot \frac{1}{hc}$$

$$= \frac{R \alpha^2 z^4}{n^3 l(l+\frac{1}{2})(l+1)} \quad \text{--- (9)}$$

α = fine structure constant.



Vector Coupling when there are more than one valence electron:

When we have more than one valence electron, say the He atom of two valence electrons, then to find the resultant angular momentum vector of the whole atom due to the two valence electrons we have to follow the coupling scheme. We use L, S, J etc to represent the quantum numbers of the whole atom. There are two possible

modes of coupling:

1. $\vec{L}-\vec{S}$ Coupling or Russel Saunders's Coupling.
2. $\vec{J}-\vec{J}$ Coupling.

Assuming the simplest case of two valence electrons, we have two different types of interaction between them.

(a) The magnetic interaction between the magnetic fields due to the orbital motion and the spinning motion of each electron.

(b) The electrostatic interaction between the charge of the two electrons.

$\vec{L}-\vec{S}$ Coupling: It is found that in lighter elements the electrostatic



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interaction is stronger than the magnetic interaction. Therefore two electron charges couple strongly to give a resultant angular momentum $\vec{P}_L$ and resultant spin quantum number $\vec{P}_S$.

$$\vec{P}_S = \vec{P}_{S_1} + \vec{P}_{S_2}$$

$$\vec{P}_L = \vec{P}_{L_1} + \vec{P}_{L_2}$$

Where $\vec{P}_{S_1}$ and $\vec{P}_{S_2}$ represent the spin angular momentum of the two electrons and $\vec{P}_{L_1}$, $\vec{P}_{L_2}$ represents the orbital angular momentum vector of the two electrons.

The total spin angular momentum vector

$$P_S = S * \hbar = \sqrt{S(S+1)} \hbar$$

Where S = the total spin quantum no. = $(S_1 + S_2)$ or $(S_1 - S_2)$.

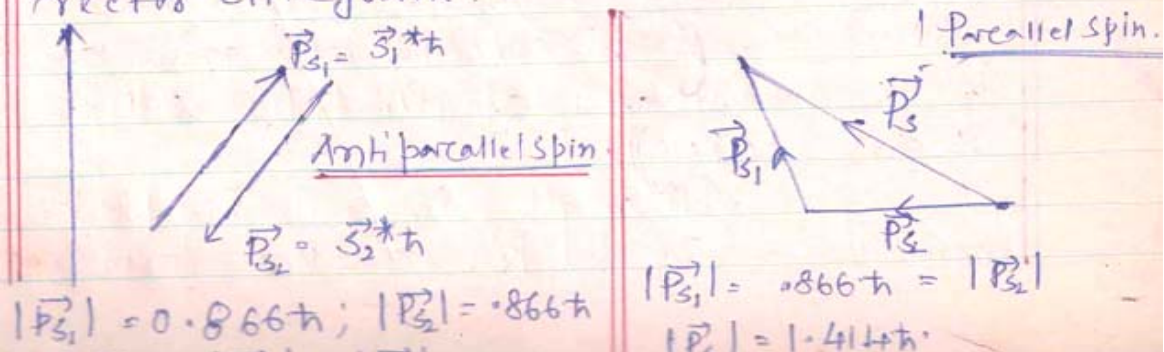
Since $S_1 = 1/2$; $S_2 = 1/2$; $S = 1$ or 0 ✓

When $S = 0$; $\vec{P}_S = 0$ ✓

When $S = 1$; $\vec{P}_S = \sqrt{1(1+1)} \hbar = \sqrt{2} \hbar = 1.414 \hbar$ ✓

$$|\vec{P}_{S_1}| = 0.866 \hbar; |\vec{P}_{S_2}| = 0.866 \hbar \quad \left[|\vec{P}_{S_1}| = \sqrt{\frac{1}{2}(\frac{1}{2}+1)} \hbar \right]$$

Let us represent the vector $\vec{P}_S$ as the resultant vectors $\vec{P}_{S_1}$ & $\vec{P}_{S_2}$ by the following vector diagram:-





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$$\vec{P}_S = \vec{P}_{S_1} + \vec{P}_{S_2}$$

$$\vec{P}_L = \vec{P}_{L_1} + \vec{P}_{L_2}$$

Where $\vec{P}_{S_1}$ and $\vec{P}_{S_2}$ represent the spin angular momentum of the two electrons and $\vec{P}_{L_1}$, $\vec{P}_{L_2}$ represents the orbital angular momentum vector of the two electrons.

The total spin angular momentum vector

$$P_S = S * h = \sqrt{S(S+1)} h$$

Where S = the total spin quantum no. = $(S_1 + S_2)$ or $(S_1 - S_2)$.

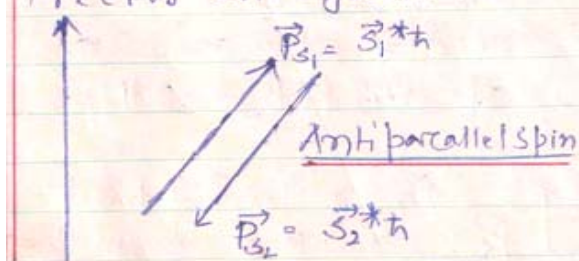
Since $S_1 = 1/2$; $S_2 = 1/2$; $S = 1$ or 0 ✓

When $S = 0$; $\vec{P}_S = 0$ ✓

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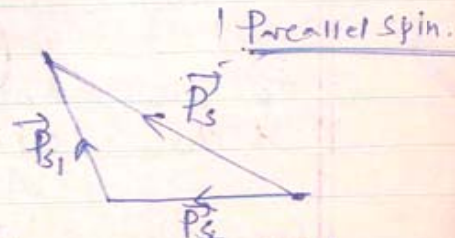
$$|\vec{P}_{S_1}| = 0.866 h; \quad |\vec{P}_{S_2}| = 0.866 h \quad \left[|\vec{P}_{S_1}| = \sqrt{\frac{1}{2}(\frac{1}{2}+1)} h \right]$$

Let us represent the vector $\vec{P}_S$ as the resultant vectors $\vec{P}_{S_1}$ & $\vec{P}_{S_2}$ by the following vector diagram:-



$$|\vec{P}_{S_1}| = 0.866 h; \quad |\vec{P}_{S_2}| = 0.866 h$$

$$\therefore |\vec{P}_S| = |\vec{P}_{S_1}| - |\vec{P}_{S_2}| = 0$$



$$|\vec{P}_{S_1}| = 0.866 h = |\vec{P}_{S_2}|$$

$$|\vec{P}_S| = 1.414 h$$



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In the presence of Strong external magnetic field, the direction of total Spin angular momentum vector $\vec{P}_S$ is also Quantised in Space; Such that the Z Component of Spin angular momentum vector is given by

$$(\vec{P}_S)_Z = m_s \hbar; \text{ where } m_s = +1 \text{ or } -1 \text{ when } S=1.$$

& $m_s = 0$, when $S=0$.

The total Orbital angular momentum vector $\vec{P}_L = L \hbar = \sqrt{L(L+1)} \hbar$

Where $L = \text{Total Orbital Quantum no.} = (l_1 + l_2) \text{ to } (l_1 - l_2) \text{ in unit steps.}$

In presence of Strong external field, the direction of total orbital angular momentum vector is also quantised in Space, such that; the Z Component of Orbital angular momentum vector is given by :-

$$(\vec{P}_L)_Z = m_L \hbar \quad \text{where } m_L = L - (L-1) \dots (L-1), L \text{ in unit steps. i.e. } m_L \text{ has } (2L+1) \text{ values.}$$

(The vector diagram of $\vec{P}_L$ as a resultant of $\vec{P}_{L1}$ & $\vec{P}_{L2}$ will be shown later.)

Although $\vec{P}_{L1}$ & $\vec{P}_{L2}$ Couple to form $\vec{P}_L$ & $\vec{P}_{S1}$ & $\vec{P}_{S2}$ Couple to form $\vec{P}_S$



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due to Strong Electrostatic interaction, but since some magnetic interaction due to the orbit and the spinning motion is still present hence $\vec{P}_L$ & $\vec{P}_S$ couple to give the total angular momentum ($\vec{P}_J$) of the atom by the relation :-

$$\vec{P}_J = \vec{P}_L + \vec{P}_S$$

$$|\vec{P}_J| = J * \hbar = \sqrt{J(J+1)} \hbar \text{ where } J \text{ is the}$$

total quantum number of the atom & is given by $J = (L+S)$ to $(L-S)$ in unit steps.

In the presence of external field the direction of the total angular momentum vector $\vec{P}_J$ is quantised in space given by

$$(\vec{P}_J)_z = m_J \hbar ; \text{ where } m_J = -J, -(J-1) \dots (J-1), J \text{ has } (2J+1) \text{ values.}$$

Let us discuss the $\vec{L}-\vec{S}$ coupling for the Helium p-d case as an example :-

Total Spin Quantum no. $S = 0$ or 1 .

Total orbital Quantum no. $L = (2+1)$ to $(2-1)$ in unit steps.
= 3, 2, 1, 0

Multiplicity of the term: $R = 2S+1$

When $S = 0$; $R = 1 \rightarrow$ Singlet. [So either we get Singlet or Triplet state]
 $S = 1$; $R = 3 \rightarrow$ Triplet



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When $L = 3$

$$l_1 = 2, |\vec{P}_{l_1}| = \sqrt{6}\hbar$$

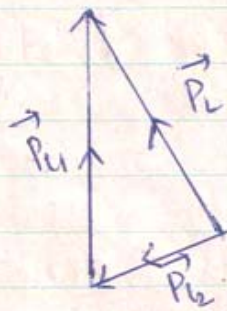
$$l_2 = 1, |\vec{P}_{l_2}| = \sqrt{2}\hbar$$

$$L = 3, |\vec{L}| = \sqrt{12}\hbar$$

$$\vec{P}_L = \vec{P}_{l_1} + \vec{P}_{l_2}$$

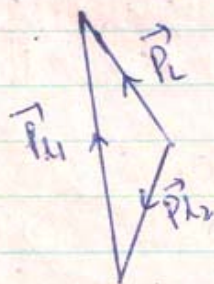


When $L = 2$, $|\vec{P}_L| = \sqrt{6}\hbar$, $|\vec{P}_{l_1}| = \sqrt{6}\hbar$, $|\vec{P}_{l_2}| = \sqrt{2}\hbar$



$$\vec{P}_L = \vec{P}_{l_1} + \vec{P}_{l_2}$$

When $L = 1$; $|\vec{P}_L| = \sqrt{2}\hbar$, $|\vec{P}_{l_1}| = \sqrt{6}\hbar$; $|\vec{P}_{l_2}| = \sqrt{2}\hbar$



$$\vec{P}_L = \vec{P}_{l_1} + \vec{P}_{l_2}$$

When $S = 0$, $J = L + S$ to $L - S$ in unit steps.

$J = L$ i.e. $J = 3, 2, 1$.

Singlet States: 1F_3 1D_2 1P_1 ($S=0, L=1$)

($S=0, L=3, J=3$), ($S=0, L=2, J=2$)



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Triplet States:- $S=1$, $R=3$

When $L=3$, $J=4, 3, 2$ ${}^3F_{4, 3, 2}$

When $L=2$; $J=3, 2, 1$ ${}^3D_{3, 2, 1}$

When $L=1$ $J=2, 1, 0$ ${}^3P_{2, 1, 0}$

J-J Coupling:-

In heavier elements, generally the magnetic interaction between the magnetic moments due to the orbital and spinning motion of the same electron is stronger than the electrostatic interaction between the charge of the electrons. In such cases, the magnetic coupling predominates the electrostatic coupling, so that orbital angular momentum vector $\vec{P}_L$ couples with spin angular momentum vector $\vec{P}_S$ of the same electron to give a resultant angular momentum vector $\vec{P}_J$ and this angular momentum vector $\vec{P}_J$ for all the valence electrons couple to give the total angular momentum vectors of the atom.

Let us consider a two valence electron atom:

Let l_1 & l_2 = Orbital Quantum number of the two electrons in the excited state.

s_1 & s_2 = Spin Quantum number ($1/2$) of the two electrons.

$$|\vec{P}_{L1}| = l_1^* \hbar; \quad |\vec{P}_{L2}| = l_2^* \hbar$$

$$|\vec{P}_{S1}| = s_1^* \hbar; \quad |\vec{P}_{S2}| = s_2^* \hbar$$



$\vec{P}_1 = \vec{P}_l + \vec{P}_{s_1}$ where the magnitude of the resultant angular momentum of the electron is $|\vec{P}_1| = j_1^* \hbar = \sqrt{j_1(j_1+1)} \hbar$

where $j_1 = (l_1 + s_1) \& (l_1 - s_1)$

Similarly for the other electron :-

$$\vec{P}_2 = \vec{P}_l + \vec{P}_{s_2}$$

$$|\vec{P}_2| = j_2^* \hbar = \sqrt{j_2(j_2+1)} \hbar$$

where $j_2 = (l_2 + s_2) \& (l_2 - s_2)$

These two angular momentum vectors $\vec{P}_1$ and $\vec{P}_2$ then couples to give the resultant angular momentum vector $\vec{P}$ given by

$$\vec{P} = \vec{P}_1 + \vec{P}_2$$

$$|\vec{P}| = J^* \hbar = \sqrt{J(J+1)} \hbar$$

where $J = (j_1 + j_2) \& (j_1 - j_2)$ in unit steps

Remark:

① It is found that J-J coupling is more important in those cases where S and P levels are inverted. Usually S level lies deeper than the P levels but in some of the heavier atoms like Si, Ge, Sn of valency 4, this sequence is inverted and the magnetic interactions seem to predominate over



Electrostatic interaction and hence J-J Coupling results.

②. The multiplicity rule is no longer holds

If there are more than two valence electrons:

L-S Coupling:

$$L = l_1 + l_2 + l_3 + \dots = \sum l \quad |\vec{P}_L| = L^* \hbar$$

$$S = s_1 + s_2 + s_3 + \dots = \sum s \quad |\vec{P}_S| = S^* \hbar$$

J = (L+S) to (L-S) in steps of unity.

$$|\vec{P}_J| = J^* \hbar = |\vec{P}_L + \vec{P}_S|$$

J-J Coupling:

$$j_1 = l_1 \pm s_1 \quad j_2 = l_2 \pm s_2 \quad j_3 = l_3 \pm s_3$$

$$J = j_1 + j_2 + j_3 + \dots = \sum j$$

$$|\vec{P}_{j_1}| = j_1^* \hbar ; \quad |\vec{P}_{j_2}| = j_2^* \hbar, \dots$$

$$|\vec{P}_J| = |\vec{P}_{j_1} + \vec{P}_{j_2} + \vec{P}_{j_3} + \dots| = J^* \hbar$$